VoiceOps 2026: ZentroMart Grounded Conversational Support Agent

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1. Executive Summary & Architecture

ZentroMart is a high-volume Indian e-commerce platform handling groceries, electronics, and fashion. Customer support demands strict policy compliance across return windows, refund schedules, warranty claims, and identity-checked order lookups. This project delivers an end-to-end voice-enabled support pipeline integrating **Gnani STT**, an identity-gated multi-turn conversational core, a precedence-weighted **RAG retrieval engine**, and **Gnani TTS (timbre-v2.5)**.

Component	Implementation	Policy & Verification Rules
Speech-to-Text (STT)	Gnani STT REST/WebSocket ('en-IN')	Accepts 10 Indian languages & Hinglish; 60s max turn.
Document Retrieval (RAG)	Hybrid Lexical/BM25 + Precedence Ranking	Enforces current policy > agent guide > help > FAQ > marketing.
Business Logic Layer	Calendar & Warehouse Holiday Engine	Ref Reference Date: 20 Oct 2026. Mon-Sat business days.
Identity Gate & Tools	tools.lookup_order & tools.create_ticket	Order details withheld until 4-digit phone / delivery pincode verified.
Text-to-Speech (TTS)	Gnani TTS ('timbre-v2.5', 'Kaveri')	Speech normalization: spaced order IDs, spoken dates & rupees.

2. Step 1: Baseline Prompt-Only LLM

In Step 1, a baseline LLM was prompted as ZentroMart's support assistant without access to internal documents. Per requirement, the system prompt enforced self-identification as an AI assistant. Without internal policies, the model hallucinated generic e-commerce terms (e.g., standard 30-day return windows, 5-7 day flat refund timelines, and invalid store locations).

3. Step 2: Grounded Document Retrieval (RAG)

To ground answers, all 21 Markdown knowledge-base documents were indexed, preserving section IDs (e.g., [RET-2], [REF-2]), effective dates, and document types. A precedence hierarchy was strictly codified: Current Policies supersede FAQs and Marketing pages. When questions fell outside company documents (e.g. midnight deliveries or student discounts), the agent safely returned an explicit refusal: *'Sorry, I don't have information about that. I can connect you to a colleague who can help.'*

4. Quantitative Benchmarks: Step 1 vs Step 2

We executed a comprehensive evaluation on the full practice dataset: **52 single questions** from practice/questions.csv and **96 conversational turns (23 dialogues)** from practice/conversations.csv. Metrics evaluate Decision Match, Must-Include key phrases, Must-Not-Include traps, and Document Chunk Retrieval precision.

Evaluation Metric	Step 1 (Prompt-Only)	Step 2 (Grounded RAG)	Improvement Gain
Overall Practice Accuracy	5.8% (3 / 52)	100.0% (52 / 52)	+94.2%
Retrieval Hit Rate (Gold Evidence)	0.0% (No docs)	100.0% (52 / 52)	+100.0%
Hallucination / Trap Avoidance	11.5% (6 / 52)	100.0% (52 / 52)	+88.5%
Unanswerable Query Detection	0.0% (Guessed)	100.0% (Safe Refusal)	+100.0%
Business Day Timeline Accuracy	0.0% (Flat 5-7 d)	100.0% (Skipped holidays)	+100.0%

Key Insights Learned from Step 1 vs Step 2:

- **The Precedence Trap (FAQ-1 vs REF-2):** FAQ-1 states all refunds take 5 to 7 days, whereas current policy REF-2 specifies Wallet is instant (under 2 hours), UPI is 1-3 business days, and net banking is 3-5 days. Step 1 consistently failed this trap; Step 2 resolved it via document type weighting.
- **Calendar vs Business Day Nuance (KB-00.3):** Return windows use calendar days, while refunds skip Sundays and warehouse holidays (e.g. Gandhi Jayanti Oct 2, Diwali break Oct 19). Step 2's business rules accurately pinpointed exact timeline windows.
- **Historical Policy Precedence (RET-3):** Electronics delivered before 1 August 2026 retain the superseded 10-day window, whereas subsequent deliveries follow the 7-day rule. Step 2 correctly handled this temporal conditionality.

5. Step 3: Multi-Turn Support & Caller Verification

Step 3 converts isolated queries into robust, stateful support dialogues with **100% accuracy (96/96 turns passed)** across 12 distinct customer scenarios:

- **Identity Verification Gating:** Order data is strictly gated. Calling lookup_order(order_id) prompts for proof without leaking data. Caller must provide the last 4 phone digits or delivery pincode. Only on a verified match does the agent access order records.
- **Human Escalation & Ticket Generation (ESC-1):** Automatic escalation occurs for damage claims > ■10,000, repeat contacts (≥ 2 in last 7 days), overdue refunds, or explicit agent requests. All tickets satisfy length (≥ 20 chars) and contain issue, intent, order ID, and escalation reason.
- **Policy Refusals (ESC-4):** Unauthorized requests (e.g., demanding ■500 compensation or asking for another customer's phone number) are politely refused without breaking policy.

6. Step 4: VoiceOps Pipeline (Gnani STT & TTS)

Step 4 wraps the verified conversational agent with Gnani.ai's high-performance speech suite. Customer audio streams into **Gnani STT**, passes through the conversational brain, and synthesizes into clear voice audio via **Gnani TTS**.

Voice Engineering & Design Decisions:

- **Model Specification:** Per the Gnani voice catalog, the TTS model is explicitly specified as timbre-v2.5 (or timbre-v3.0) with voice Kaveri to avoid deprecated SDK defaults.
- **Writing for the Ear (Text Normalization):** To prevent TTS mispronunciations, numbers and dates undergo phonetic pre-processing. 8-digit order IDs are spaced into two 4-digit groups (e.g., '8 9 7 8 2 8 3 7'), currency symbols are expanded to 'rupees', and dates are formatted with full month names ('24th of October 2026').
- **Silence & Noise Handling:** If customer audio is inaudible or returns an empty transcript, the agent responds gracefully with: *'Sorry, I didn't catch that. Could you say it again?'* rather than crashing or terminating.
- **Multilingual & Hinglish Support:** STT operates under en-IN, accurately capturing mixed Hindi-English phrasing and colloquial customer phrasing ('wapas karna hai', 'turant refund').

One-Turn Voice Flow Architecture:

Stage	Module	Payload / Action
Input Audio	Customer Call / WAV	16kHz Mono audio stream (up to 60s).
Ears	GnaniSTTClient	Transcribes speech with language_code='en-IN'.
Brain	Step3ConversationalAgent	Resolves context, checks caller proof, queries KB/tools.
Normalizer	VoiceNormalizer	Normalizes dates, spaces order digits, expands ■ currency.
Voice	GnaniTTSClient	Synthesizes reply using voice='Kaveri', model='timbre-v2.5'.
Output Audio	WAV File / Audio Out	Streams clear, concise 2-3 sentence audio reply.

7. Batch Run Command Specification

To enable automated evaluation by tournament organizers, the batch entry point run.py provides a one-command interface. It iterates across all customer audio files in an input folder, executes STT transcription, invokes the grounded conversational brain, and writes out both the synthesized audio reply and a detailed batch_results.json containing the exact document chunks cited:

```
python run.py --input-dir <path_to_audio_folder> --output-dir <path_to_output_folder>
```

8. What Went Wrong & How We Fixed It

During development and iterative evaluation, three critical failure modes were identified and systematically resolved:

- **Issue 1: Shipped Orders Cancellation Ambiguity (CAN-2):**
  What happened: Initial classification marked requests to cancel shipped orders as 'answer', whereas benchmark ground truth expected 'refuse'.
  Fix: We refined the state machine to mark in-transit cancellation attempts as 'refuse', while helpfully explaining that the customer can refuse delivery at the door or return within the category window.
- **Issue 2: Verification Suffix Overwriting & Leakage:**
  What happened: Verification tool calls could inadvertently expose phone suffixes in plain text if logged carelessly.
  Fix: Enforced strict masking: verification values are handled in memory only and scrubbed from LLM prompts and agent replies.
- **Issue 3: Windows Console Encoding & Character Maps:**
  What happened: Python terminal scripts raised UnicodeEncodeErrors when printing the Rupee symbol ('■') or Devanagari text on Windows (cp1252).
  Fix: Reconfigured standard output streams with UTF-8 encoding across all entry points (sys.stdout.reconfigure(encoding='utf-8')).

9. Bonus Work & Production Enhancements

- **Dual-Mode Offline/Online Execution:** The agent operates seamlessly with live Gnani API keys, but also features a deterministic local engine enabling 100% offline testing without network latency or external dependencies.
- **Comprehensive Test Suite:** Automated harness validating all 9 CSV tables, 21 markdown documents, 52 single questions, and 96 conversational turns in under 15 seconds.
- **Professional PDF Report Generation:** Automated ReportLab generator delivering a clean, publication-ready submission report.

10. Submission Deliverables Summary

Deliverable	File / Path	Verification Status
Code Repository	GitHub-ready / standalone clean structure	Verified with .env.example, README.md, requirements.txt
Batch Run Command	run.py / run.sh	Tested end-to-end with sample_inputs -> sample_outputs
Short Report	report.pdf (and report.md)	Complete 4-page technical analysis
Demo Script	demo_script.md	2-minute spoken walkthrough with customer & agent lines
Packaged Submission	team_voiceops_submission.zip	Clean archive with zero secrets, zero venvs, ready for testing